

Seaborn

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
```

```
sns.set_theme(style='whitegrid') # styles: whitegrid, darkgrid, white, dark, ticks
```

```
# Load the Titanic dataset (built-in!)
df = sns.load_dataset('titanic')
print(df.head())
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

Heatmap — Correlations between columns

showing data using colours

```
import seaborn as sns
import matplotlib.pyplot as plt

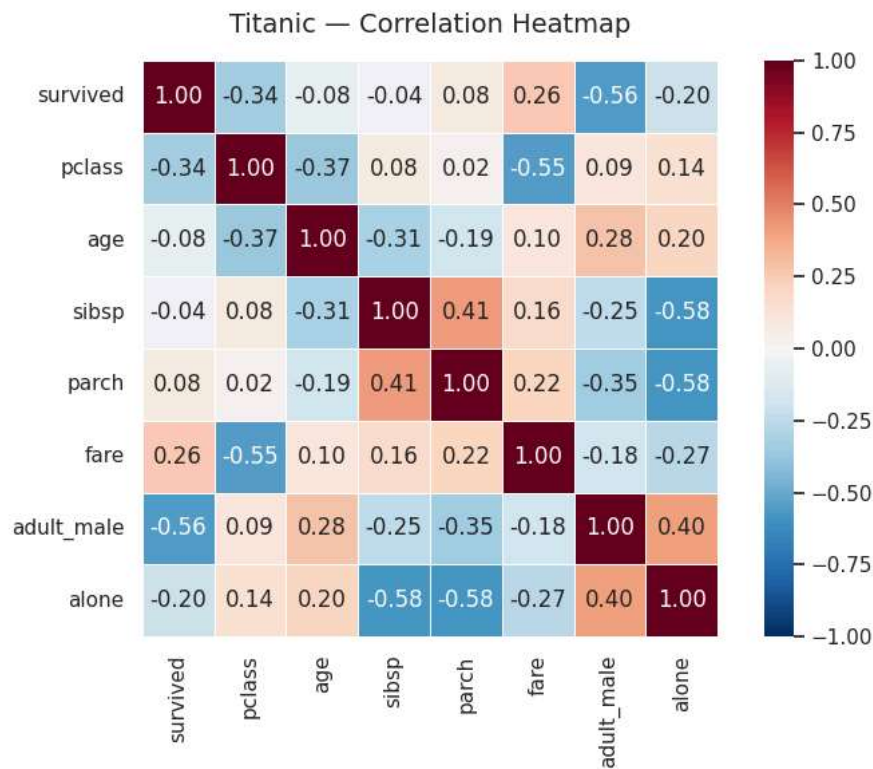
# Example: Titanic dataset (or your df)
# df = pd.read_csv("titanic.csv")

corr_matrix = df.corr(numeric_only=True)

plt.figure(figsize=(8, 6))

sns.heatmap(corr_matrix,
            annot=True,
            fmt='.2f',
            cmap='RdBu_r',
            vmin=-1, vmax=1,
            linewidths=0.5,
            square=True)

plt.title('Titanic - Correlation Heatmap', fontsize=14, pad=15)
plt.tight_layout()
plt.savefig('heatmap.png', dpi=150, bbox_inches='tight')
plt.show()
```



Boxplot — Spotting Outliers

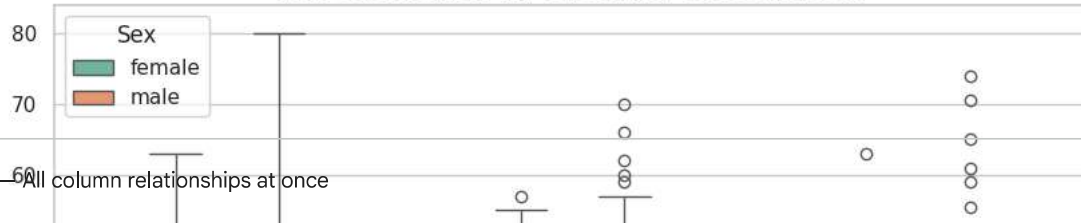
```
# Anatomy of a boxplot:
# Box top    = 75th percentile (Q3)
# Box bottom = 25th percentile (Q1)
# Line       = Median (Q2)
# Whiskers   = Q1 - 1.5*IQR to Q3 + 1.5*IQR
# Dots       = Outliers beyond whiskers

plt.figure(figsize=(10, 6))
sns.boxplot(data=df,
            x='pclass',          # 3 groups: 1st, 2nd, 3rd class
            y='age',              # numerical variable
            hue='sex',            # split by gender (2 colours)
            palette='Set2',
            width=0.6)

plt.title('Age Distribution by Passenger Class and Sex', fontsize=14)
plt.xlabel('Passenger Class')
plt.ylabel('Age')
plt.legend(title='Sex')
plt.savefig('boxplot.png', dpi=150, bbox_inches='tight')
plt.show()

# What to look for:
# - Dots = outliers (very old or very young passengers)
# - Box size = spread (3rd class has wider age range)
# - Median line position = skewness
```

Age Distribution by Passenger Class and Sex



Pairplot — All column relationships at once

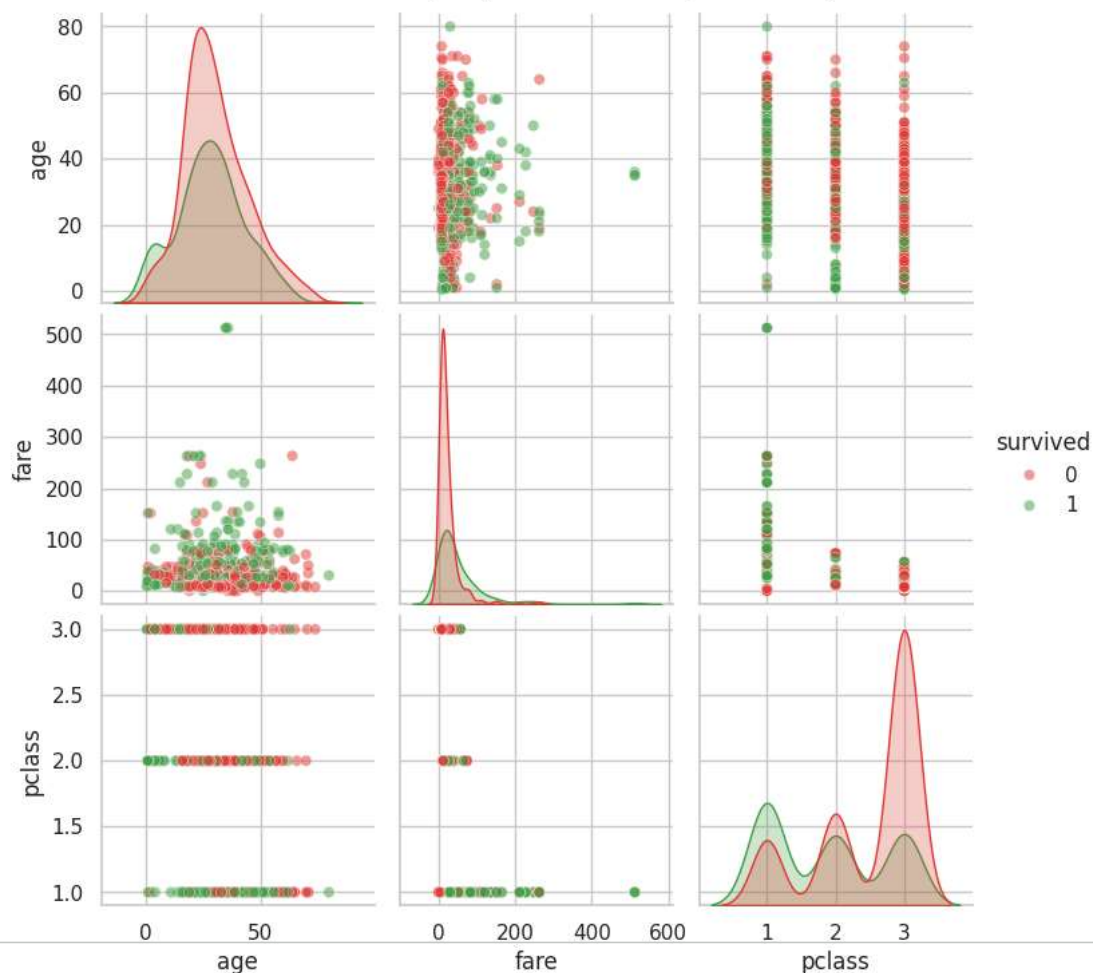
```
df_sub = df[['age', 'fare', 'survived', 'pclass']].dropna()

pairplot = sns.pairplot(df_sub,
                        hue='survived',          # colour by survival (0 or 1)
                        palette={0: '#E53935', 1: '#43A047'},
                        diag_kind='kde',          # diagonal: density curves
                        plot_kws={'alpha': 0.5},
                        height=2.5)
```

```
pairplot.fig.suptitle('Titanic — Pairplot (Green=Survived, Red=Died)',
                      y=1.02, fontsize=13)
pairplot.savefig('pairplot.png', dpi=150, bbox_inches='tight')
plt.show()
```

```
# Reading the pairplot:
# - Top row: age vs fare, age vs pclass, age vs survived
# - Diagonal: distribution of that single variable
# - Green clusters separate from red = good predictor of survival
```

Titanic — Pairplot (Green=Survived, Red=Died)



sns.set_theme(style=) Set style

sns.heatmap(corr) Heatmap

sns.boxplot(x=y,hue=) Boxplot

sns.pairplot(df) All pairs

annot=True Show values

cmap='RdBu_r' Diverging colours